CLASS 10 — SCIENCE · NCERT-TYPE MCQ TEST SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. In myopia, the image of a distant object is formed 1

(a) behind the retina **(b)** on the retina
(c) in front of the retina **(d)** on the iris

2. The V-I graph for an ohmic conductor is 1

(a) a parabola **(b)** a straight line through the origin
(c) a horizontal line **(d)** a curve

3. An electromagnet is made of 1

(a) copper only
(b) soft iron wrapped with a coil carrying current
(c) steel only
(d) plastic

4. The various steps in a food chain are called 1

(a) habitats **(b)** ecosystems **(c)** trophic levels **(d)** niches

5. In the electrolysis of water, the volumes of hydrogen and oxygen collected are in the ratio: 1

(a) 2 : 1 **(b)** 1 : 1 **(c)** 3 : 1 **(d)** 1 : 2

6. Methyl orange in acidic solution is: 1

(a) colourless **(b)** yellow **(c)** red **(d)** blue

7. In the electrolytic refining of copper, the electrolyte is: 1

(a) molten copper **(b)** dilute HCl
(c) NaCl solution **(d)** acidified copper sulphate solution

8. The number of isomers of butane is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

9. The filtration of blood in the nephron takes place in: 1

(a) collecting duct **(b)** glomerulus **(c)** tubule **(d)** ureter

10. Dwarfism is caused by a deficiency of: 1

(a) insulin **(b)** adrenaline **(c)** growth hormone **(d)** thyroxine

11. The ova (eggs) are produced in 1

(a) oviduct **(b)** uterus **(c)** ovaries **(d)** vagina

12. A normal human male has the sex chromosomes 1

(a) XY **(b)** XX **(c)** YY **(d)** XXY

13. SI unit of power of a lens is 1

(a) dioptre **(b)** metre **(c)** watt **(d)** newton

14. Which colour bends most on passing through a prism? 1

(a) Yellow **(b)** Red **(c)** Violet **(d)** Orange

15. In a parallel circuit the quantity that is the same across all the resistors is 1

(a) current **(b)** resistance

(c) potential difference **(d)** power

16. An induced current is produced in a coil when 1

(a) a magnet is held still near it

(b) a magnet is moved towards or away from it

(c) a battery is disconnected

(d) the coil is kept in a steady field at rest

17. The ozone layer protects the earth from 1

(a) harmful ultraviolet radiation **(b)** sound waves

(c) visible light **(d)** infrared rays

18. Magnesium ribbon is cleaned before burning in air in order to: 1

(a) make it burn with a green flame

(b) remove the protective oxide layer from its surface

(c) make it heavier

(d) convert it into a powder

19. Which type of medicine is used for treating indigestion? 1

(a) antibiotic **(b)** analgesic **(c)** antiseptic **(d)** antacid

20. Food cans are coated with tin and not with zinc because: 1

(a) zinc has a higher melting point than tin **(b)** zinc is more reactive than tin

(c) zinc is less reactive than tin **(d)** zinc is cheaper than tin

21. The next homologue of C_3H_6 is: 1

(a) C_5H_{10} **(b)** C_4H_8 **(c)** C_4H_{10} **(d)** C_3H_8

22. The breakdown of pyruvate to give carbon dioxide, water and energy takes place in: 1

(a) nucleus **(b)** mitochondria **(c)** chloroplast **(d)** cytoplasm

23. The hormone which helps in the growth of the shoot tip towards light is: 1

(a) abscisic acid **(b)** auxin **(c)** cytokinin **(d)** gibberellin

24. Rhizopus reproduces by 1

(a) spores **(b)** buds **(c)** binary fission **(d)** fragmentation

25. When pure tall (TT) and pure dwarf (tt) pea plants are crossed, the F1 generation is 1

(a) all tall **(b)** all dwarf

(c) three tall to one dwarf **(d)** half tall and half dwarf

26. The focal length of a spherical mirror of radius of curvature R is 1

(a) R/2 **(b)** R/4 **(c)** 2R **(d)** R

27. The ability of the eye lens to adjust its focal length is called 1

(a) myopia **(b)** dispersion **(c)** scattering **(d)** accommodation

28. Potential difference is measured by a 1

(a) galvanometer in series **(b)** ammeter in parallel

(c) voltmeter connected in series **(d)** voltmeter connected in parallel

29. The strength of a magnetic field is indicated by 1

(a) the closeness of the field lines **(b)** the number of poles

(c) the colour of the lines **(d)** the length of the lines

30. Organisms that depend on others for food are called 1

(a) decomposers only **(b)** autotrophs **(c)** consumers **(d)** producers

31. In $\mathrm{CuO} + \mathrm{H_2} \longrightarrow \mathrm{Cu} + \mathrm{H_2O}$, the substance oxidised is: 1

(a) CuO **(b)** Cu **(c)** $\mathrm{H_2O}$ **(d)** $\mathrm{H_2}$

32. The sting of a bee contains: 1

(a) lactic acid **(b)** ethanoic acid **(c)** methanoic acid **(d)** citric acid

33. Which of the following is an ionic compound? 1

(a) $\mathrm{H_2O}$ **(b)** $\mathrm{MgCl_2}$ **(c)** $\mathrm{CCl_4}$ **(d)** $\mathrm{CH_4}$

34. Esters are formed when a carboxylic acid reacts with: 1

(a) an alkane **(b)** a metal **(c)** an alcohol **(d)** a base

35. The energy currency of the cell is: 1

(a) ATP **(b)** glucose **(c)** starch **(d)** ADP

36. Growth of the pollen tube towards the ovule is an example of: 1

(a) geotropism **(b)** thigmotropism **(c)** chemotropism **(d)** phototropism

37. Transfer of pollen from anther to stigma is called 1

(a) fertilisation **(b)** fission **(c)** pollination **(d)** germination

38. The new combinations of traits (round green and wrinkled yellow) in F2 show that 1

(a) no dominant trait exists **(b)** F1 gets only one parent's traits

(c) traits are always linked **(d)** traits are inherited independently

39. If light enters a medium of refractive index 1.5 from vacuum, its speed becomes (c = 3×10^8 m/s) 1

(a) 4.5×10^8 m/s **(b)** 1.5×10^8 m/s **(c)** 3×10^8 m/s **(d)** 2×10^8 m/s

40. In hypermetropia, the near point is 1

(a) farther than 25 cm **(b)** nearer than 25 cm

(c) at infinity **(d)** at 25 cm

41. The resistance of a conductor is 1

(a) inversely proportional to its length **(b)** proportional to the square of the area

(c) directly proportional to its length **(d)** independent of length

42. The strength of an electromagnet can be increased by 1

(a) decreasing the current
(b) increasing the current or the number of turns
(c) using a thicker iron-free coil
(d) removing the core

43. In a food chain, the first trophic level is occupied by 1

(a) carnivores **(b)** producers **(c)** decomposers **(d)** herbivores

44. $CaCO_3 \longrightarrow CaO + CO_2$ is an example of: 1

(a) electrolytic decomposition **(b)** displacement
(c) photochemical decomposition **(d)** thermal decomposition

45. The pH of pure water at room temperature is: 1

(a) 7 **(b)** 14 **(c)** 5 **(d)** 0

46. Brass is an alloy of: 1

(a) copper and tin **(b)** copper and zinc **(c)** iron and carbon **(d)** lead and tin

47. Diamond is a: 1

(a) good conductor **(b)** liquid
(c) non-conductor and very hard **(d)** soft solid

48. Artificial kidney works on the principle of: 1

(a) dialysis **(b)** filtration only **(c)** osmosis only **(d)** photosynthesis

49. Diabetes is caused due to deficiency of: 1

(a) insulin **(b)** adrenaline **(c)** oestrogen **(d)** thyroxine

50. Fertilisation in human females normally occurs in the 1

(a) ovary **(b)** vagina
(c) oviduct (fallopian tube) **(d)** uterus

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND